Daniyal Kashif

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EDUCATION

University of Arizona Tucson, Arizona
Bachelor of Science, Computer Science - Minor in AI, Minor in Game Development Expected May 2027
Dean's List (Dec 2023 – Present) - Academic Distinction
Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Computer Organization, Software Development, Linear Algebra, Principles of Data Science, Computational Linguistics, Systems Programming and Unix

Skills

Languages: Python, Java, JavaScript, TypeScript, HTML/CSS, C, SQL
ML/Data: Scikit-learn, XGBoost, PyTorch, Pandas, NumPy, NLTK, Streamlit, FastAPI
Web & Tools: Next.js, React, Django, PostgreSQL, Redis, Prisma, Vite, Clerk, Git, Figma

Work Experience

FBM Solutions Remote, USA
AI/ML Intern *June 2025-Aug 2025*

- Developed and deployed an intelligent chatbot system using FastAPI, integrating document retrieval (RAG) with external API data sources.
- Fine-tuned DistilBERT transformer model for intent classification, achieving multi-class prediction across policy and data retrieval intents.
- Integrated external REST APIs with JWT authentication for real-time employee data retrieval and leave balance queries.
- Implemented hybrid routing system with keyword-based pre-filtering to optimize query processing.

University of Arizona Tucson, Arizona
Research Assistant *November 2023-September 2024*

- Annotated and evaluated 500+ paragraphs using Human Signal and Label Studio, identifying key discrepancies between human and AI language understanding to support experimental NLP research.
- Collaborated with NLP researchers to compare differences in human vs. AI annotations across multi-subject sentence interpretation tasks.
- Developed and implemented experimental strategies to address identified challenges, emphasizing a hands-on approach to problem-solving.
- Investigated key limitations of large language models in parsing multi-verb or multi-subject inputs to support future model improvements.

Technical Projects

Watchlist TV Recommender | source code | *Next.js · React · Python · Prisma · Tailwind CSS*

- Built full-stack TV recommendation app with personalised user experiences using Next.js, React, and Python.
- Integrated Prisma ORM with TMDB API to fetch and manage show data across personalised recommendation pages.
- Built dynamic, reusable React components and implemented routing across 5+ pages in a collaborative team environment.

Credit Card Fraud Detector | source code | *Python · XGBoost · SMOTE · Streamlit*

- Trained XGBoost classifier on 6.3M transactions achieving 0.9992 ROC-AUC and 0.9984 AUC-PR — 25x precision improvement over baseline logistic regression.
- Engineered domain-specific fraud features (account drain patterns, balance mismatches) and applied SMOTE to handle 0.13% class imbalance.
- Tuned decision threshold to optimise F1, catching 1,640 of 1,643 fraud cases in test set. Deployed live on Streamlit Cloud.

Fridge Inventory Web App | source code | *React · Django · PostgreSQL · Redis · Celery*

- Built full-stack web app with React frontend and Django REST Framework backend to manage food inventory and expiration dates.
- Configured Celery with Redis for background task processing and automated email reminders for expiring items.
- Implemented JWT authentication and designed intuitive UI components for adding, updating, and deleting inventory items.

Online Sexism Detector | source code | *Python · Scikit-learn · TF-IDF · SVM · Streamlit*

- Built NLP classifier achieving 0.81 weighted F1 on the EDOS benchmark dataset across 4,193 labelled examples.
- Benchmarked 6 model/feature combinations (Logistic Regression, SVM, XGBoost × Word/Char/Combined TF-IDF) to identify optimal pipeline.
- Combined word-level and character-level TF-IDF to capture both semantic patterns and misspelling/slang signals common in online toxic language.

Publications

- Kashif, D., [et al.]. *"BEMEAE: Moving Beyond Exact Span Match for Event Argument Extraction."* In **Proceedings of the NAACL 2025 Main Conference**. ACL Anthology, 2025. https://aclanthology.org/2025.naacl-long.295/